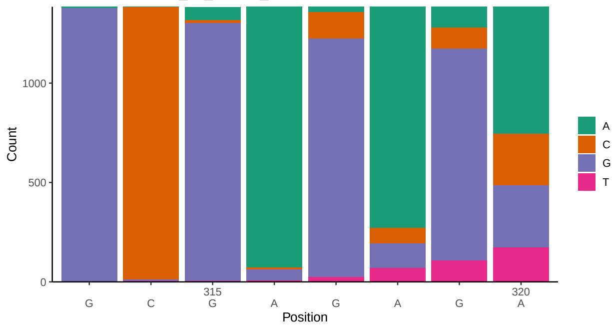
OGRDBstats Report

Contents

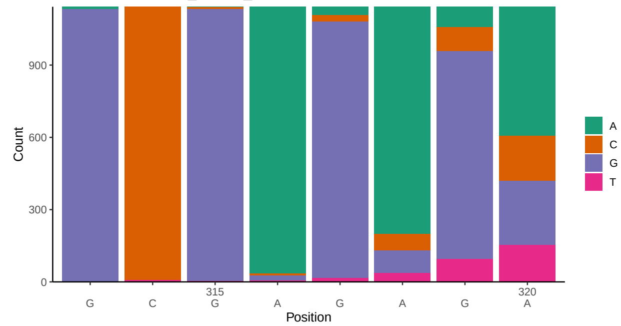
1 Novel sequence analysis

1.1 End-nucleotide composition

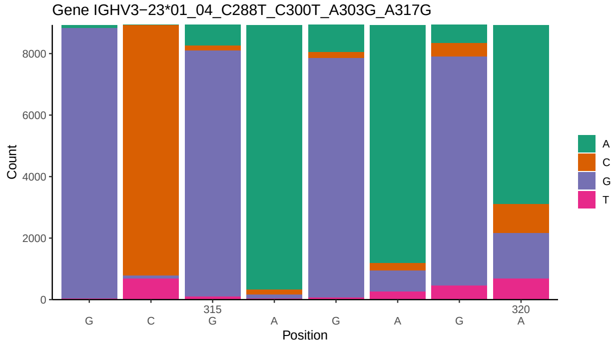
Gene IGHV1−3*01_05_A291G_T300C



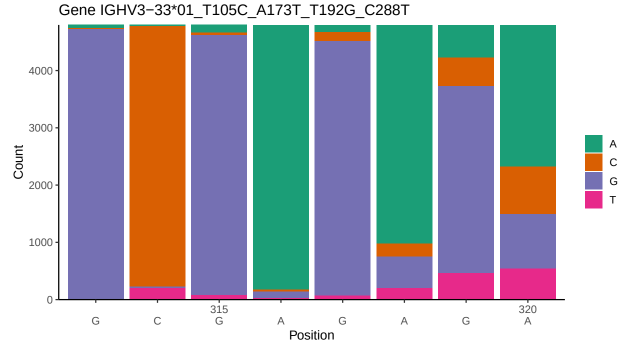
Gene IGHV1−24*01_A315G_C317G



Gene IGHV3−23*01_04_C288T_C300T_A303G_A317G



Gene IGHV3−33*01_T105C_A173T_T192G_C288T



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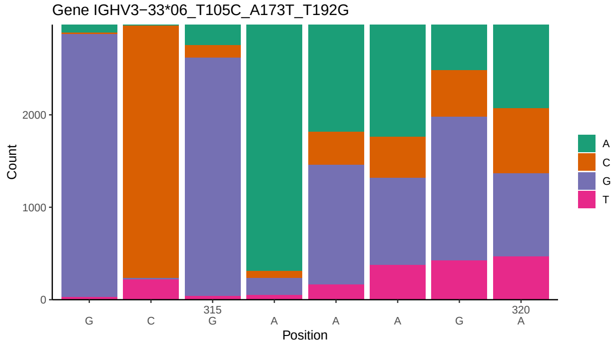

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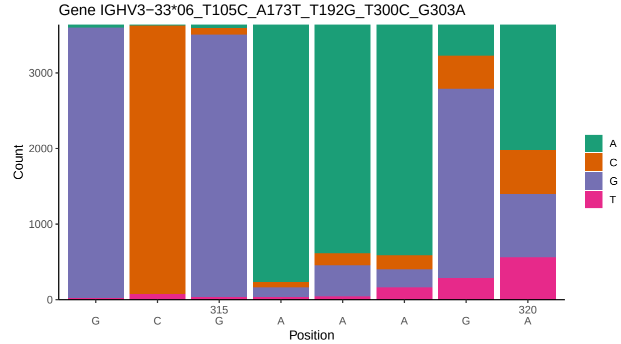
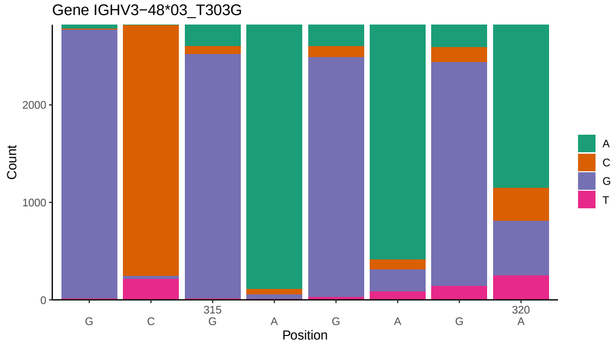
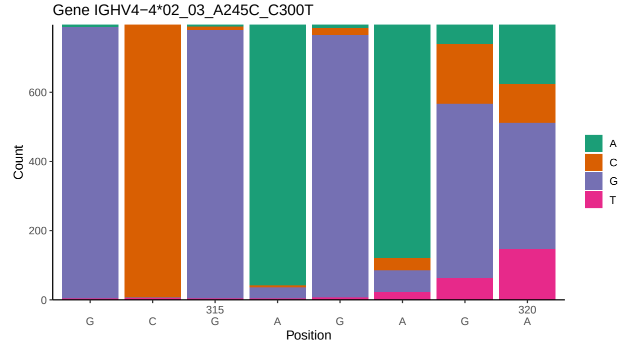
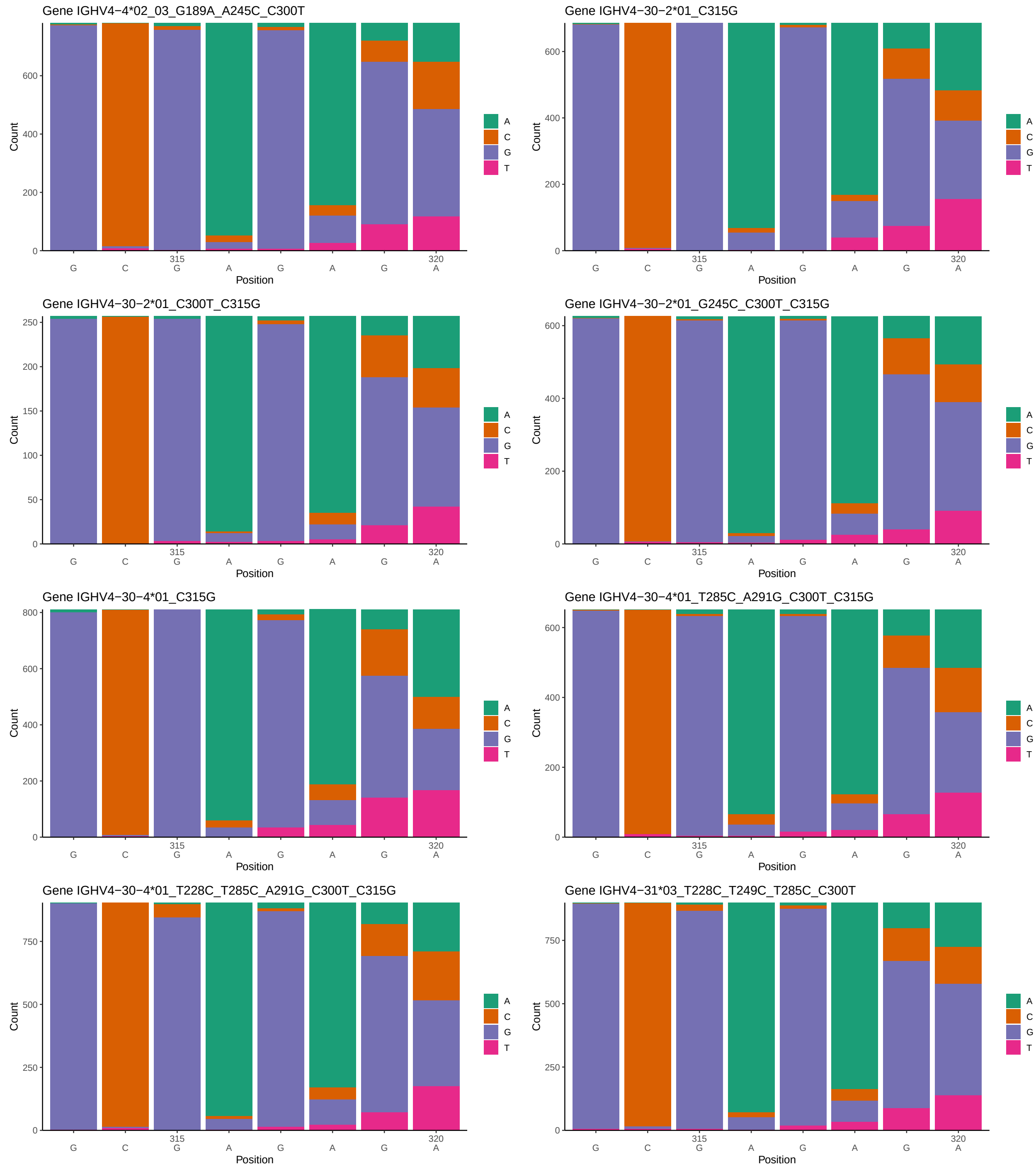


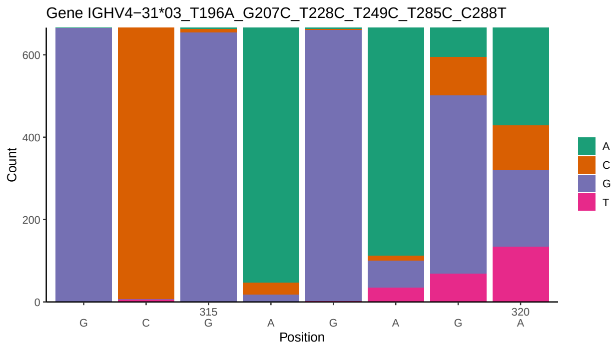
Gene IGHV3−48*03_T303G

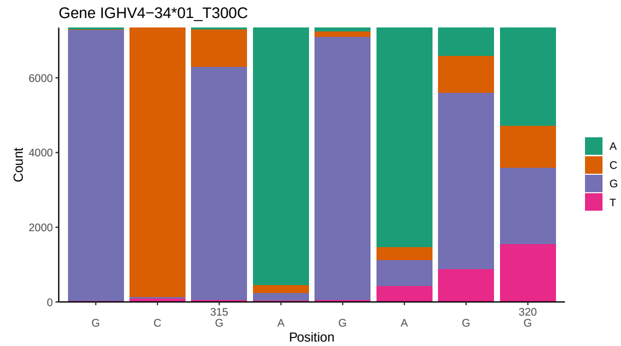


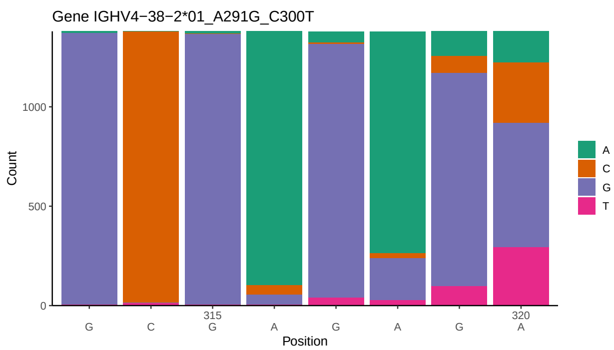
Gene IGHV4−4*02_03_A245C_C300T

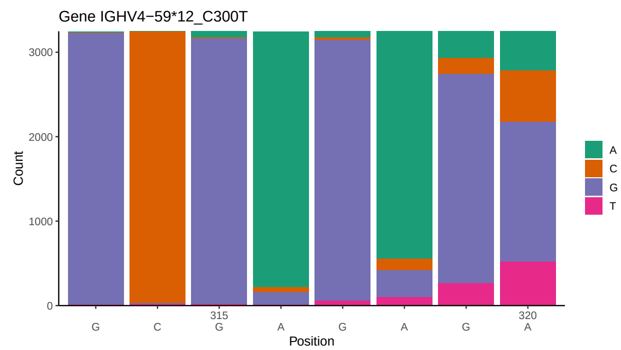



Gene IGHV4−4*02_03_G189A_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−2*01_C300T_C315G
Gene IGHV4−30−2*01_G245C_C300T_C315G
Gene IGHV4−30−4*01_C315G
Gene IGHV4−30−4*01_T285C_A291G_C300T_C315G
Gene IGHV4−30−4*01_T228C_T285C_A291G_C300T_C315G
Gene IGHV4−31*03_T228C_T249C_T285C_C300T
Count
Position
G C 315 G A G A G 320 A
A
C
G
T

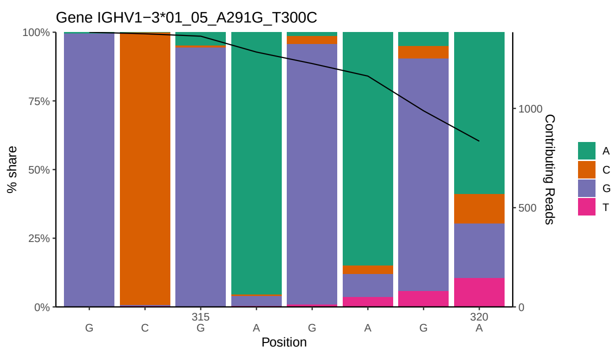


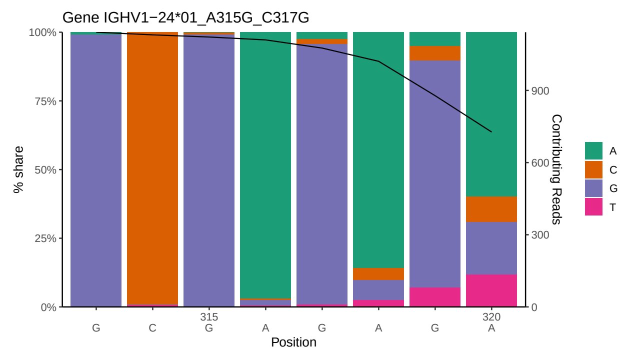


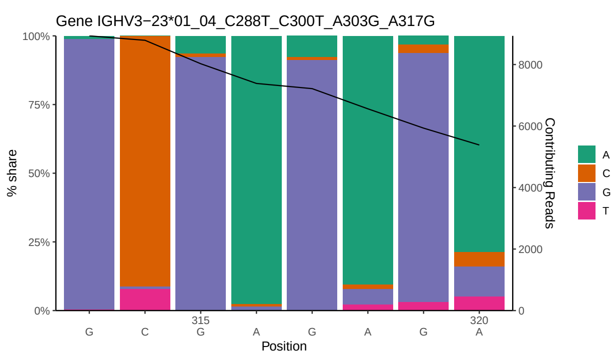


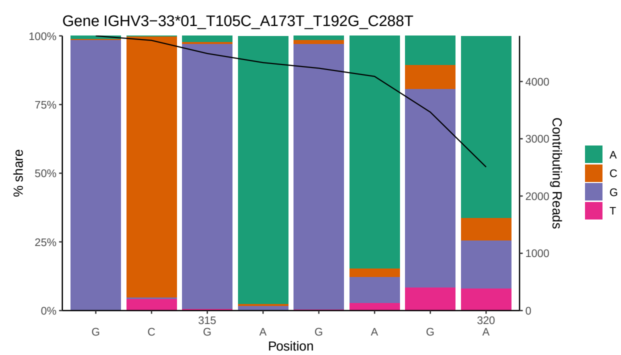


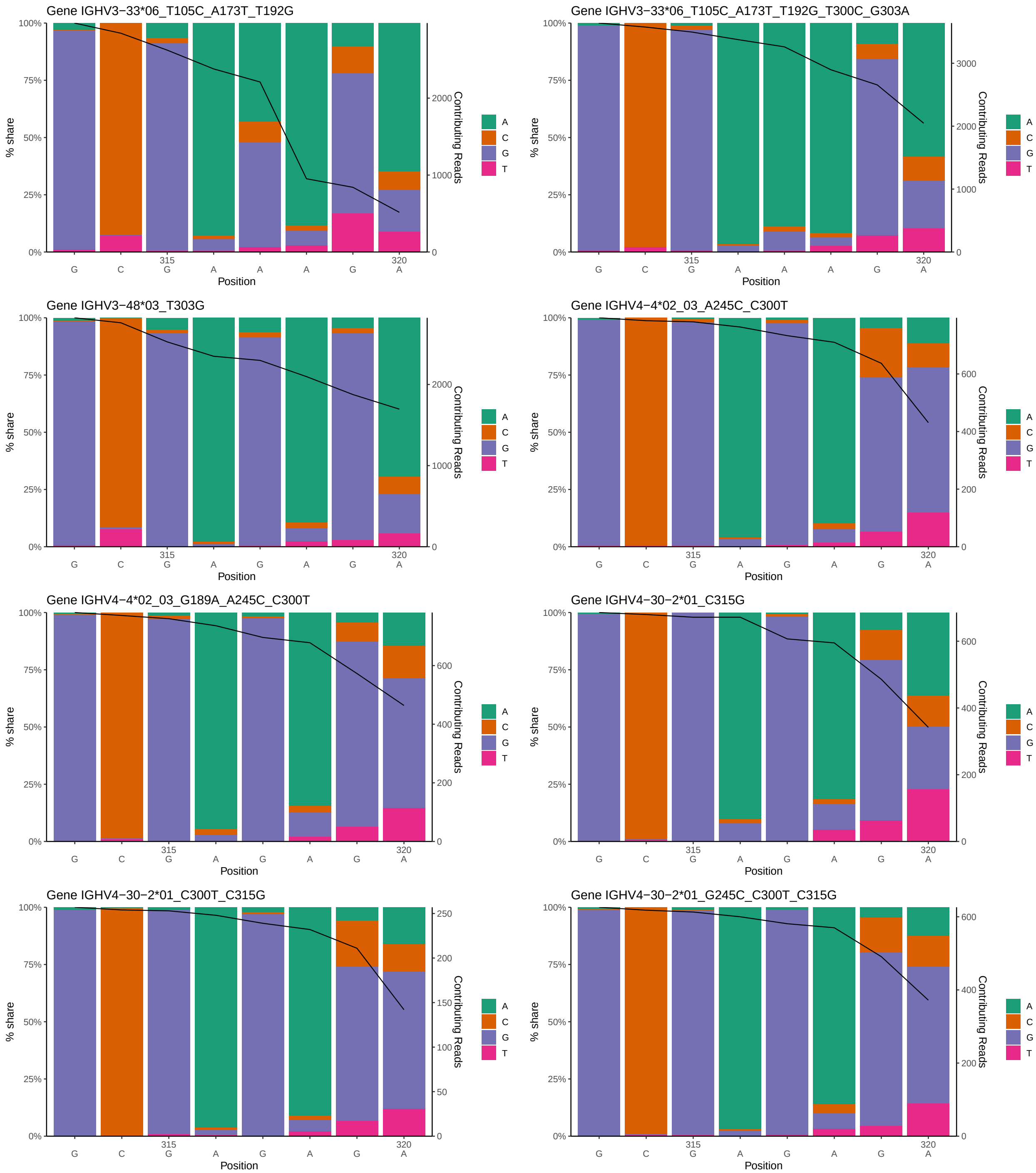
1.2 Per-nucleotide consensus where previous nucleotides match the consensus



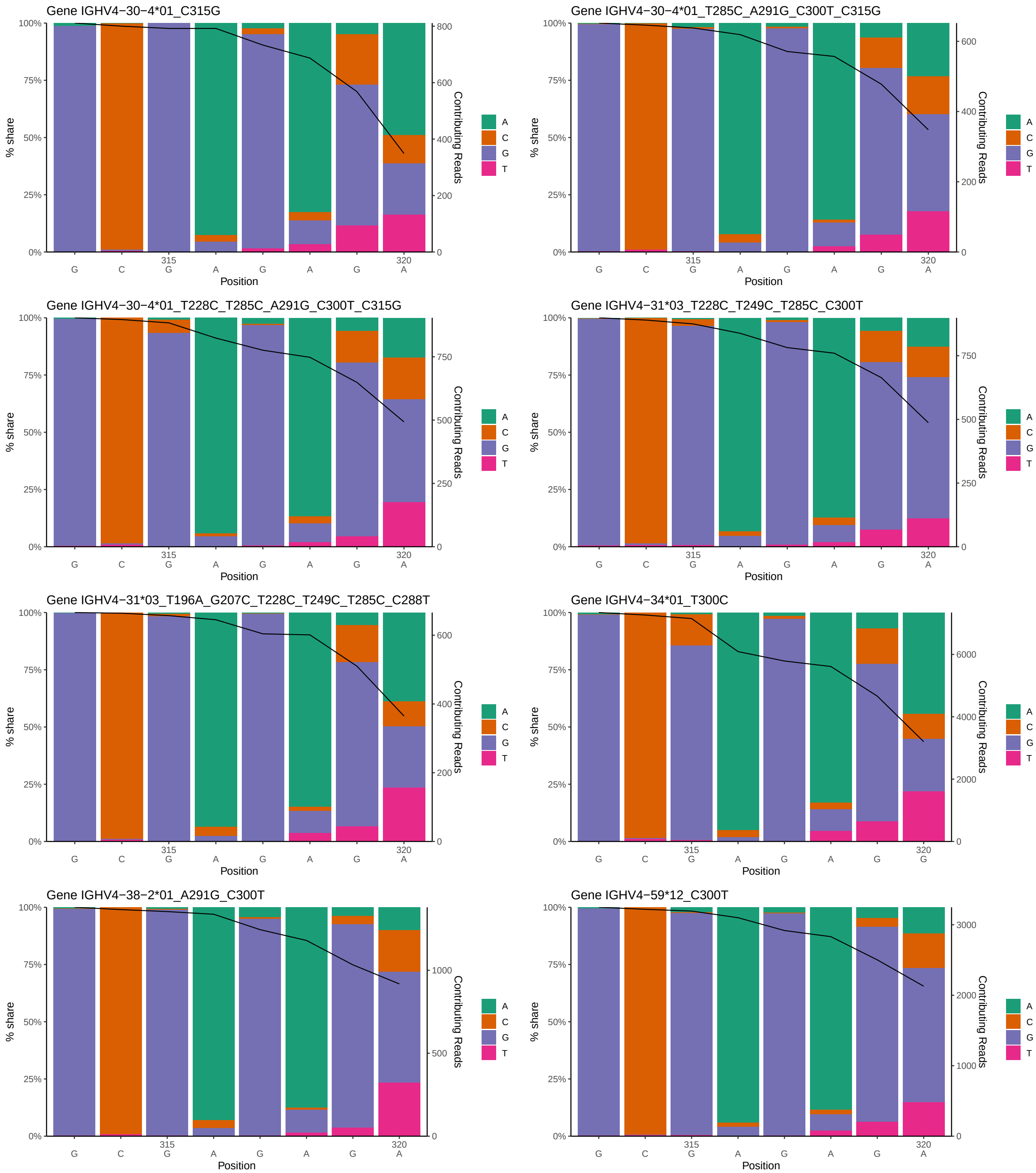








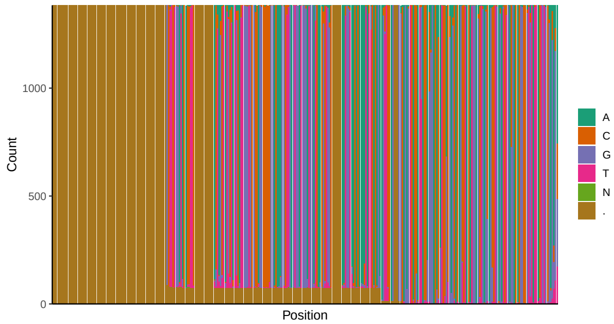
Gene IGHV3−33*06_T105C_A173T_T192G
Gene IGHV3−33*06_T105C_A173T_T192G_T300C_G303A
Gene IGHV3−48*03_T303G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−4*02_03_G189A_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−2*01_C300T_C315G
Gene IGHV4−30−2*01_G245C_C300T_C315G
% share
Contributing Reads
Position
A
C
G
T



Gene IGHV4−30−4*01_C315G
Gene IGHV4−30−4*01_T285C_A291G_C300T_C315G
Gene IGHV4−30−4*01_T228C_T285C_A291G_C300T_C315G
Gene IGHV4−31*03_T228C_T249C_T285C_C300T
Gene IGHV4−31*03_T196A_G207C_T228C_T249C_T285C_C288T
Gene IGHV4−34*01_T300C
Gene IGHV4−38−2*01_A291G_C300T
Gene IGHV4−59*12_C300T
% share
Contributing Reads
Position
A
C
G
T

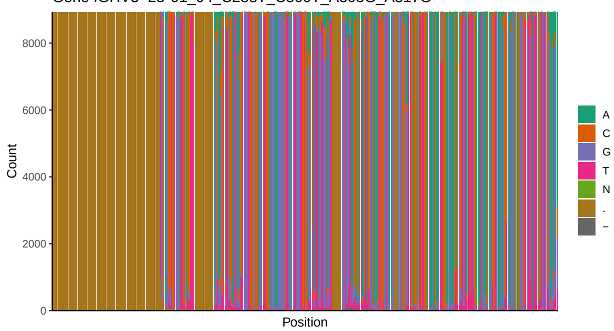
1.3 Whole-sequence composition of each assigned read

Gene IGHV1−3*01_05_A291G_T300C

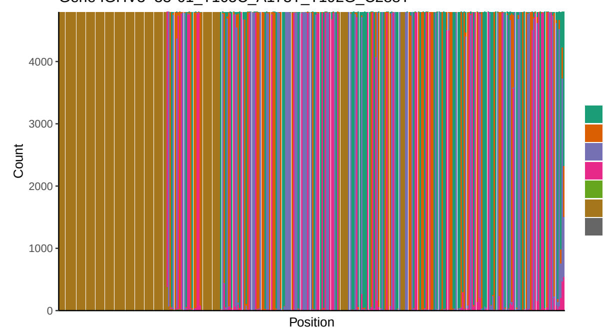


Gene IGHV1−24*01_A315G_C317G

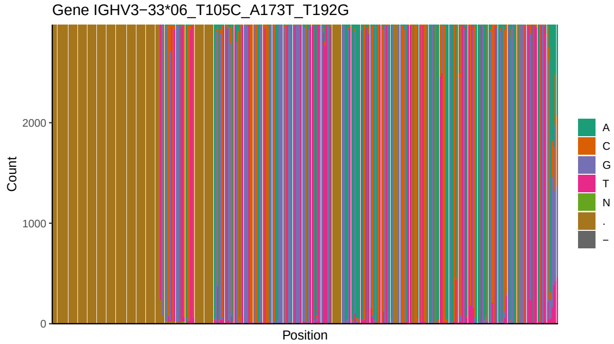
Gene IGHV3−23*01_04_C288T_C300T_A303G_A317G



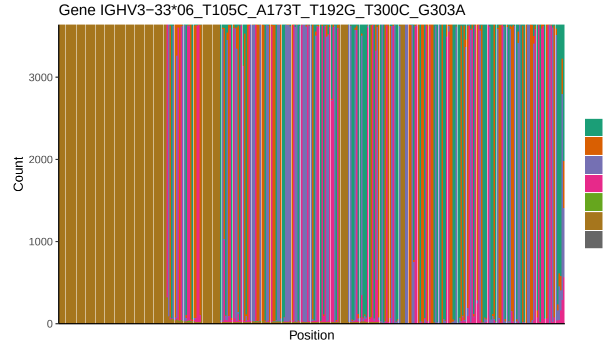
Gene IGHV3−33*01_T105C_A173T_T192G_C288T



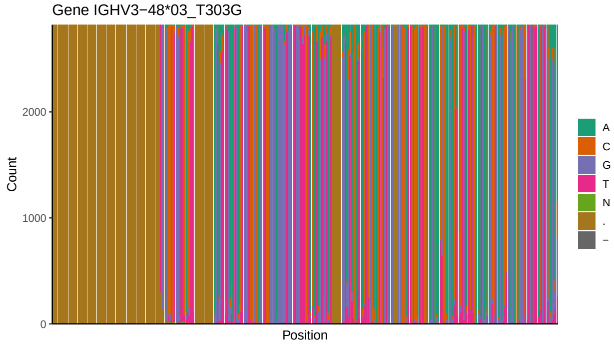
Gene IGHV3−33*06_T105C_A173T_T192G



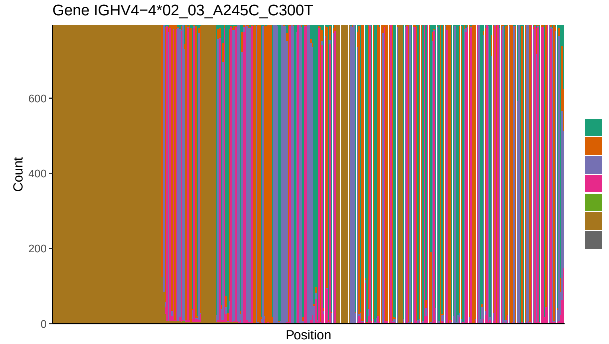
Gene IGHV3−33*06_T105C_A173T_T192G_T300C_G303A

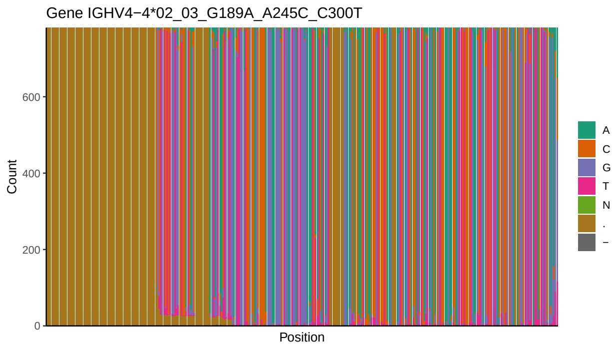


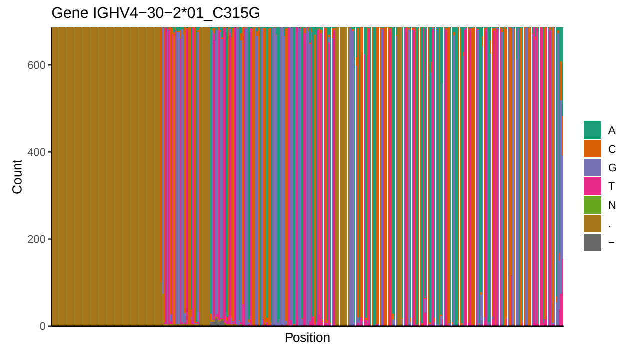
Gene IGHV3−48*03_T303G

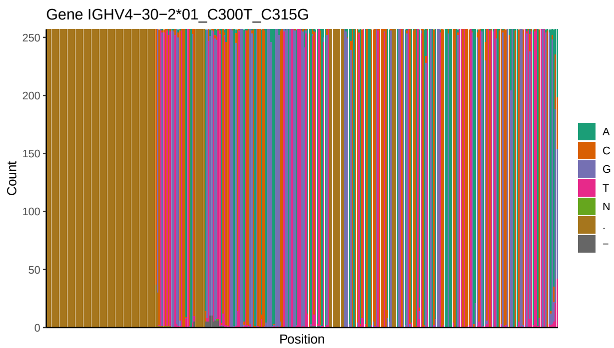


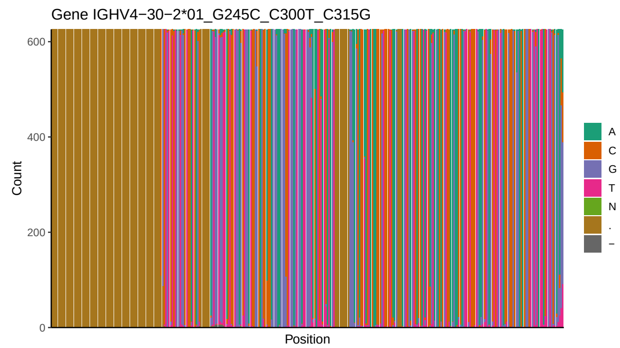
Gene IGHV4−4*02_03_A245C_C300T

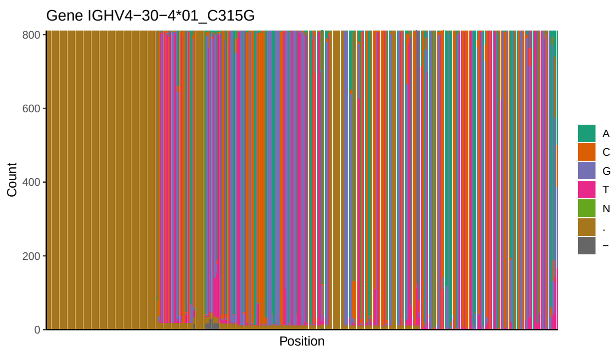


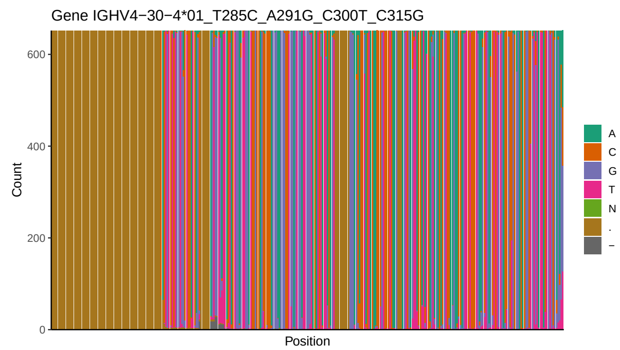

Gene IGHV4−4*02_03_G189A_A245C_C300T
Count
Position
A
C
G
T
N
.
−

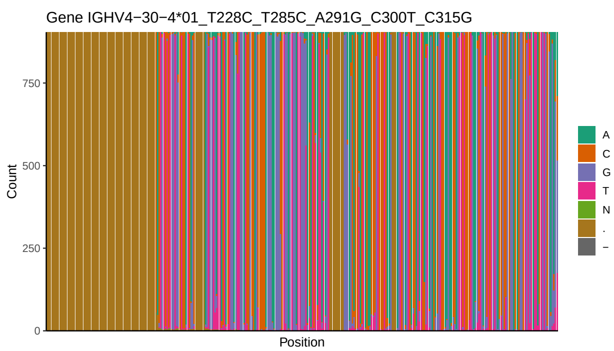

Gene IGHV4−30−2*01_C315G
Count
Position
A
C
G
T
N
.
−

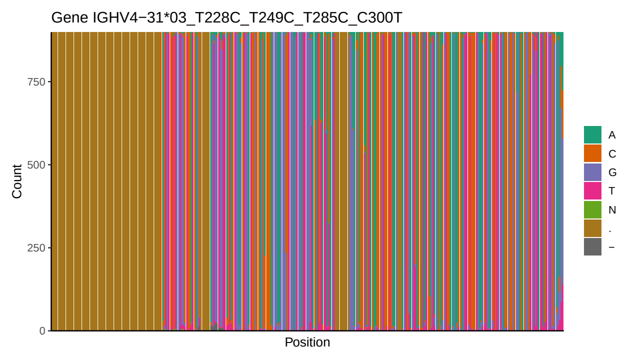

Gene IGHV4−30−2*01_C300T_C315G
Count
Position
A
C
G
T
N
.
−

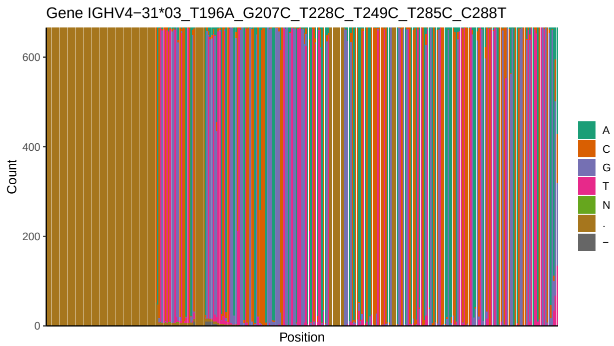

Gene IGHV4−30−2*01_G245C_C300T_C315G
Count
Position
A
C
G
T
N
.
−

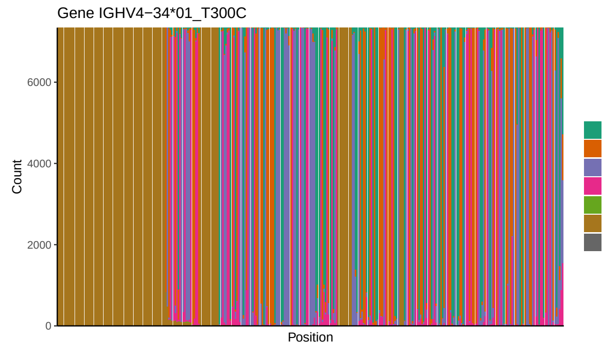

Gene IGHV4−30−4*01_C315G
Count
Position
A
C
G
T
N
.
−

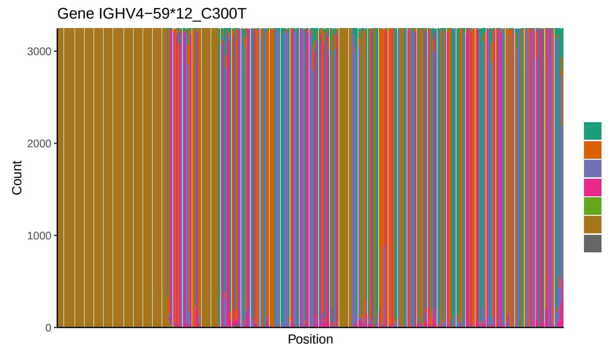

Gene IGHV4−30−4*01_T285C_A291G_C300T_C315G
Count
Position
A
C
G
T
N
.
−


Gene IGHV4−30−4*01_T228C_T285C_A291G_C300T_C315G
Count
Position
A
C
G
T
N
.
−

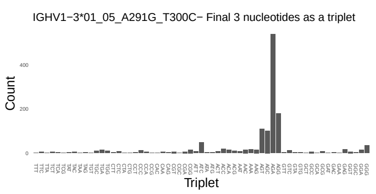

Gene IGHV4−31*03_T228C_T249C_T285C_C300T
Count
Position
A
C
G
T
N
.
−

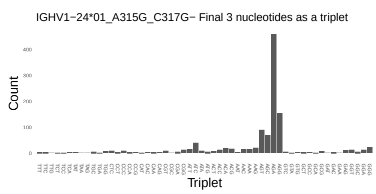


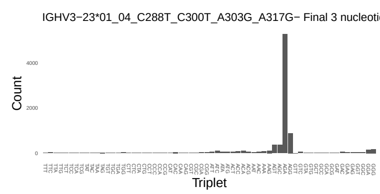


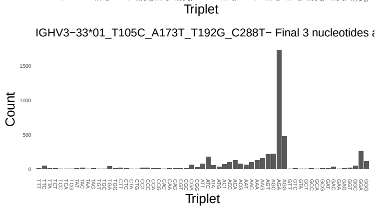


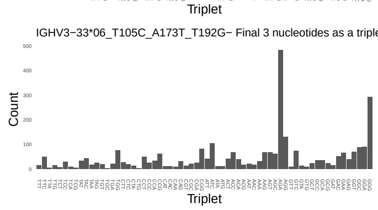
1.4 Final three nucleotides: frequency of each observed triplet

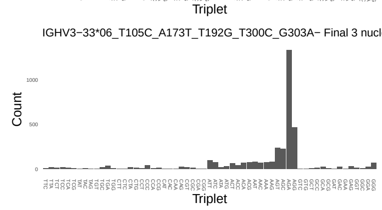


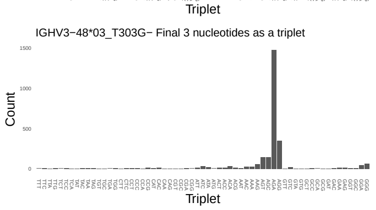


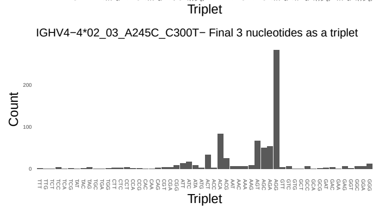


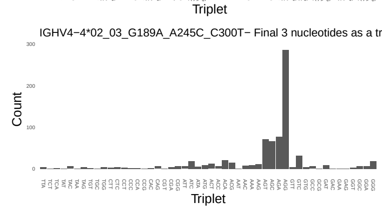


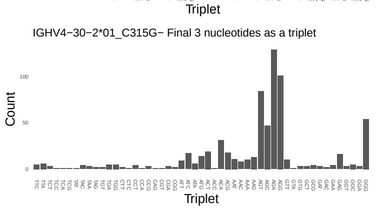


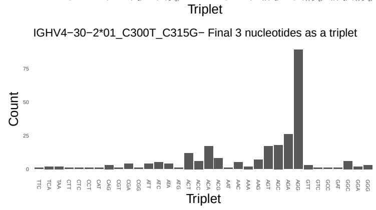


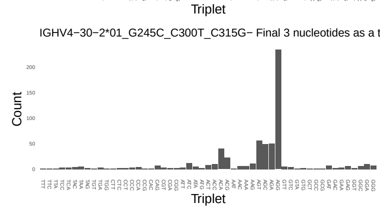


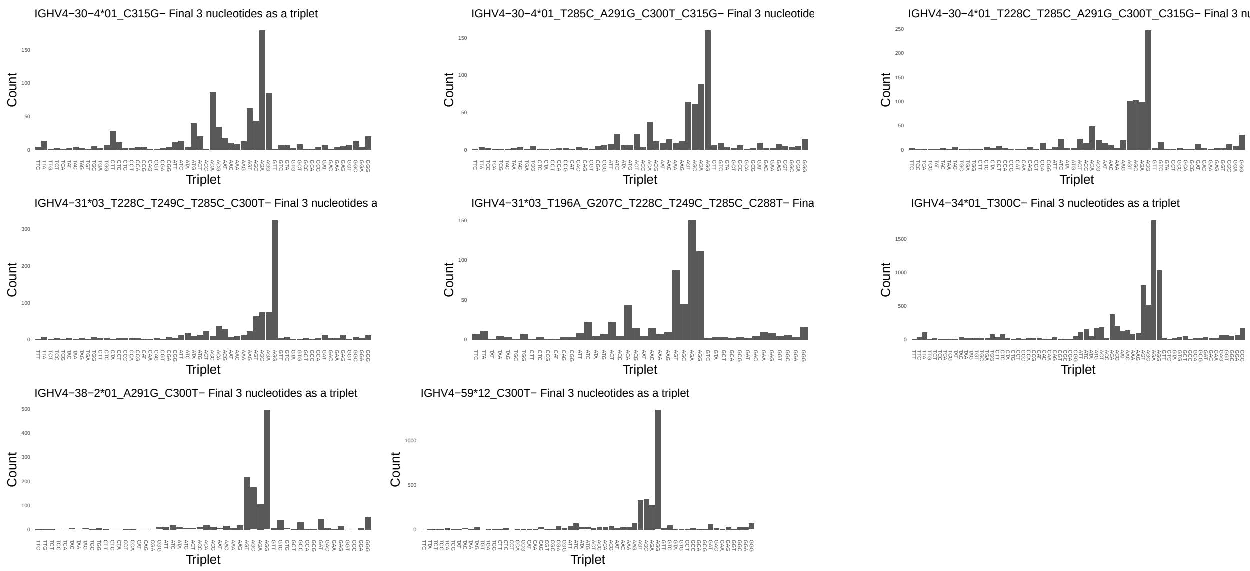




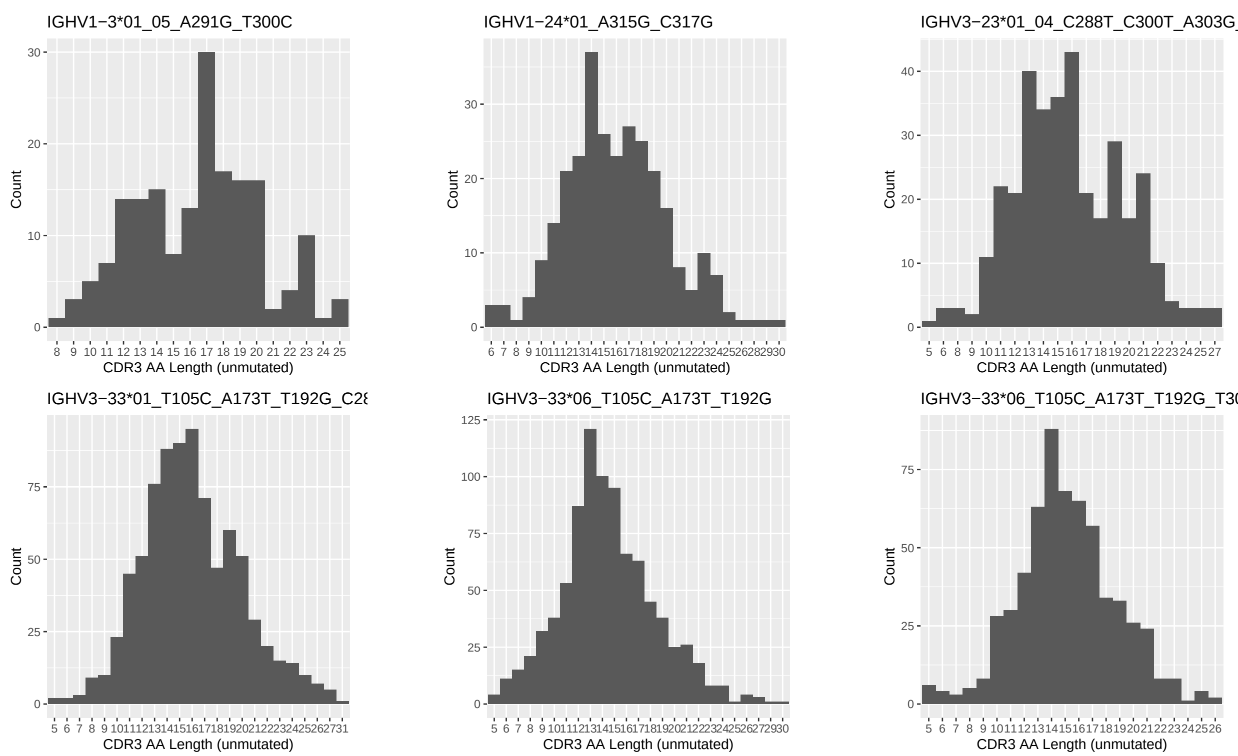


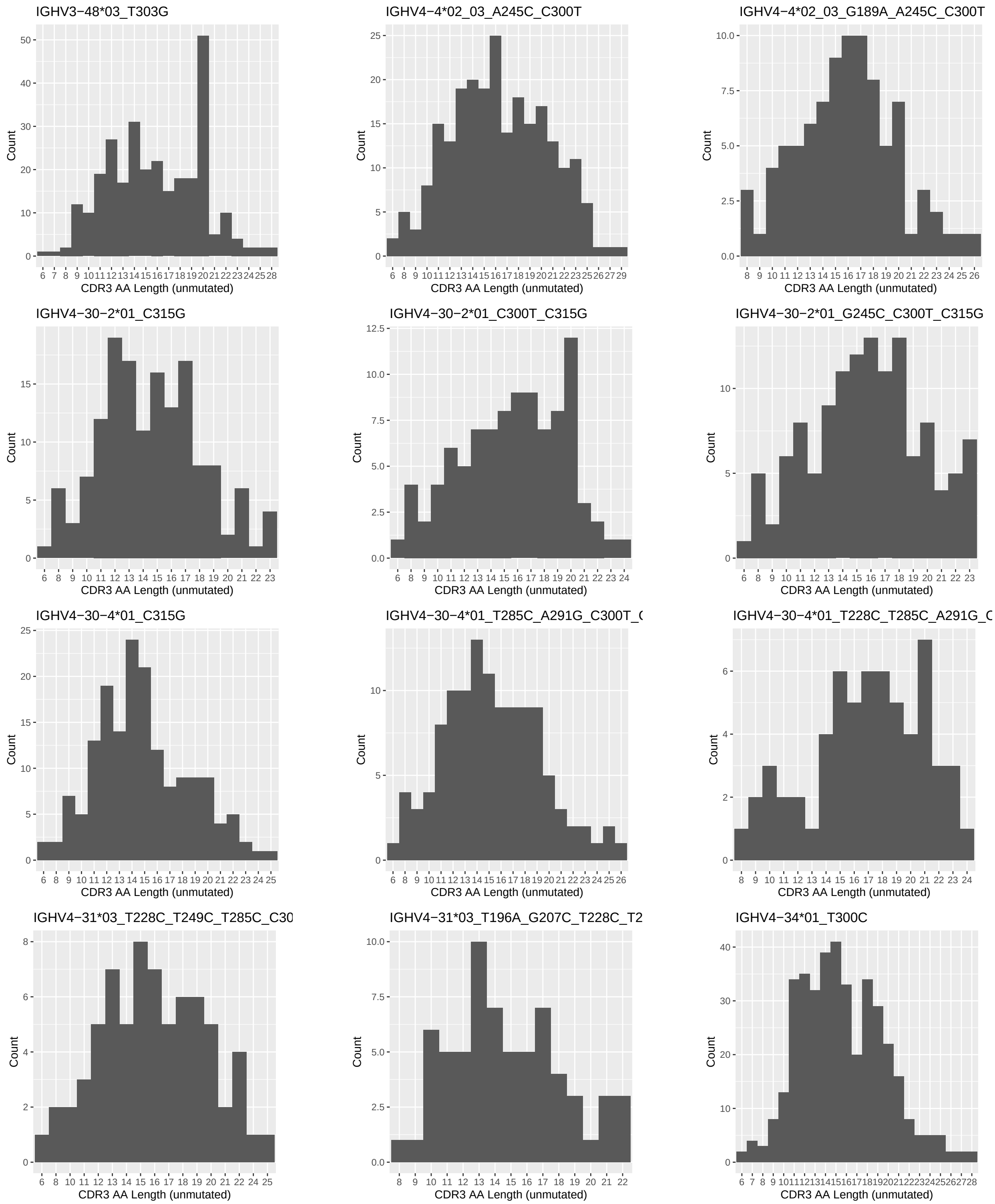


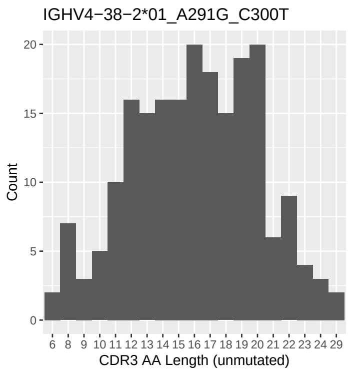


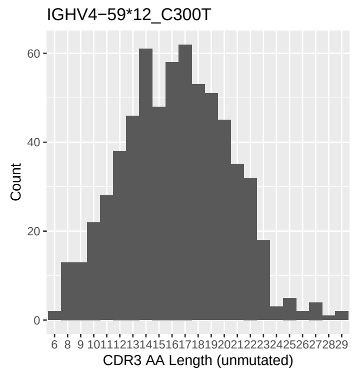


1.5 CDR3 length distribution, in assignments to novel alleles

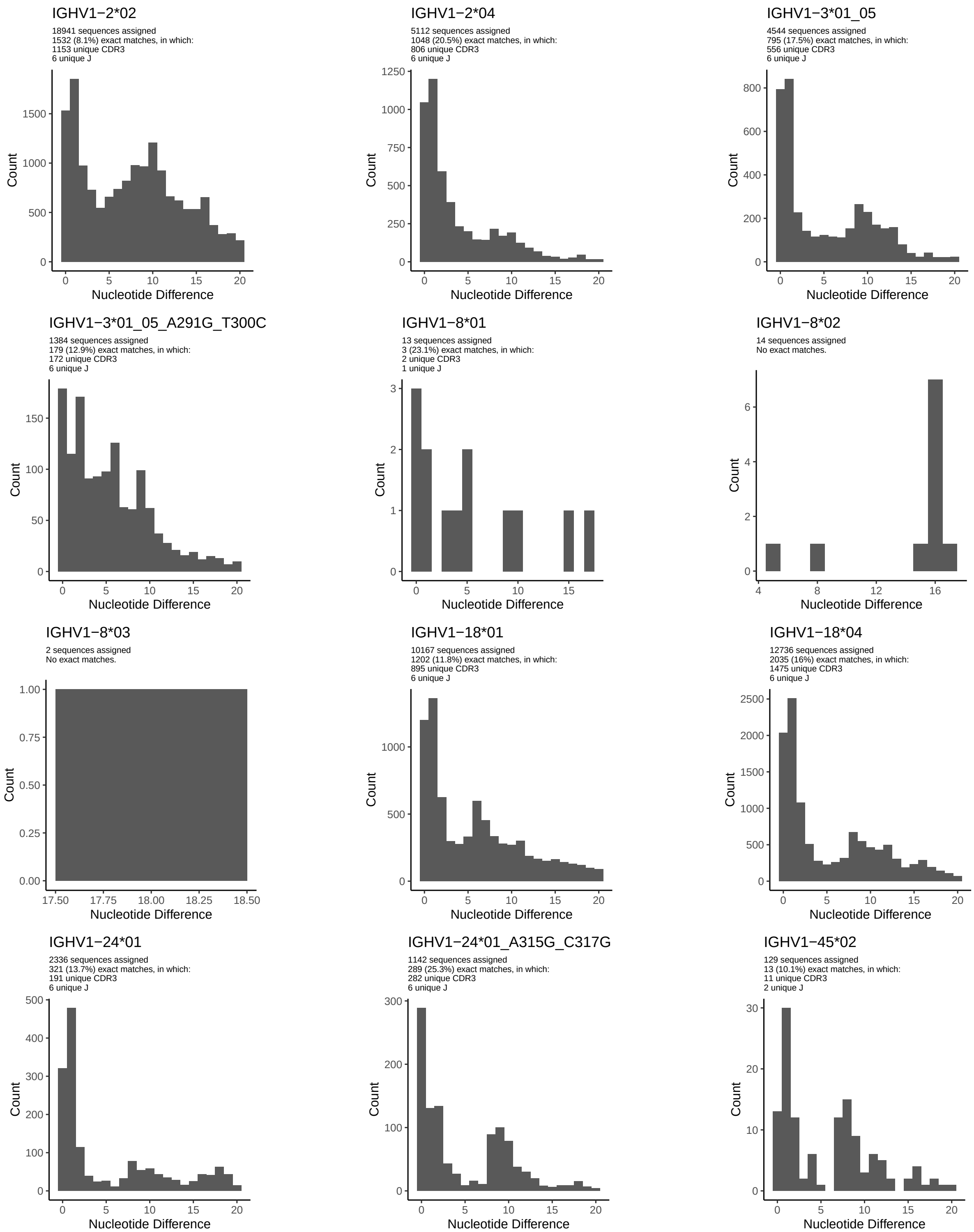


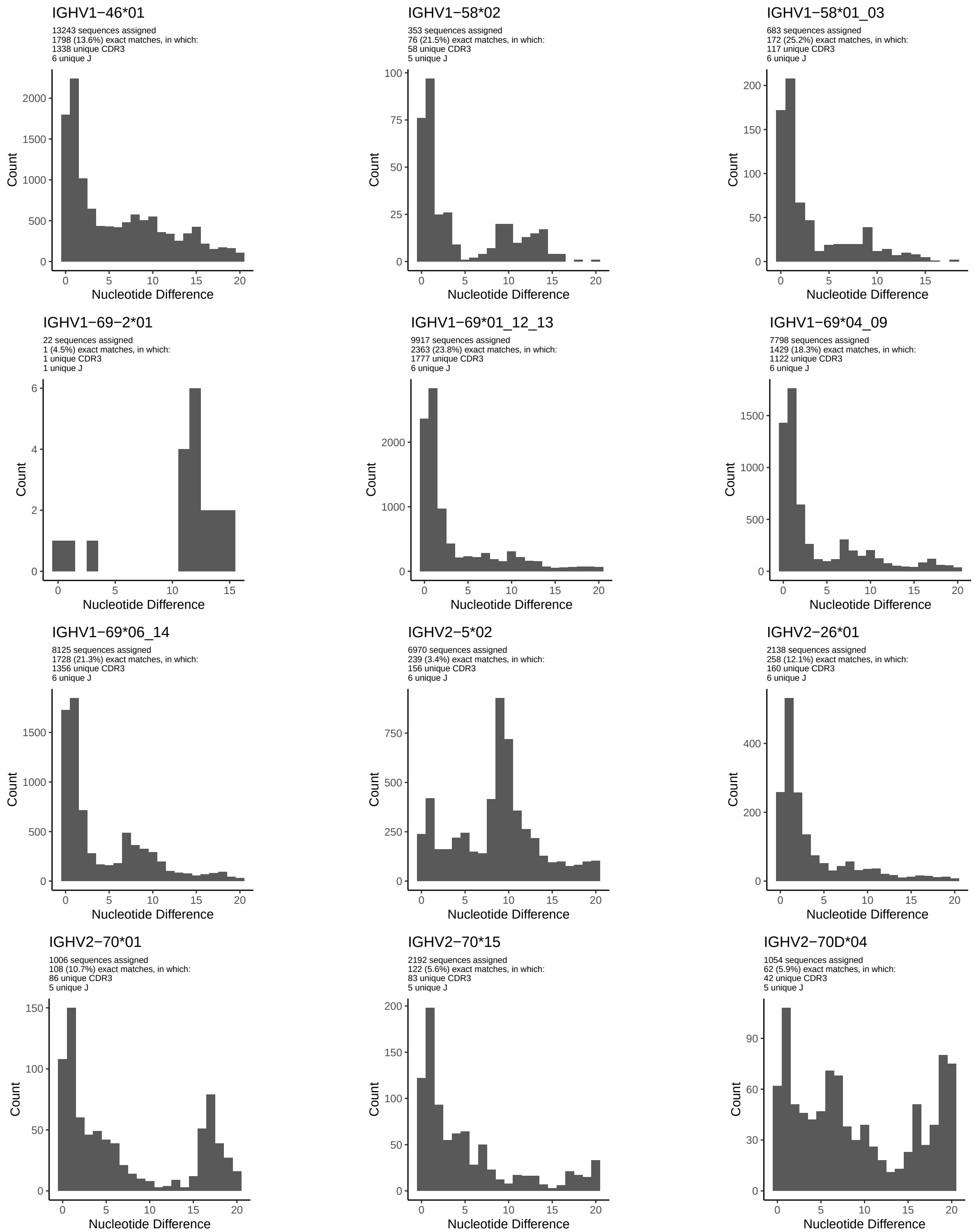

IGHV3−48*03_T303G
IGHV4−4*02_03_A245C_C300T
IGHV4−4*02_03_G189A_A245C_C300T
IGHV4−30−2*01_C315G
IGHV4−30−2*01_C300T_C315G
IGHV4−30−2*01_G245C_C300T_C315G
IGHV4−30−4*01_C315G
IGHV4−30−4*01_T285C_A291G_C300T_C
IGHV4−30−4*01_T228C_T285C_A291G_C
IGHV4−31*03_T228C_T249C_T285C_C30
IGHV4−31*03_T196A_G207C_T228C_T2
IGHV4−34*01_T300C
Count
CDR3 AA Length (unmutated)

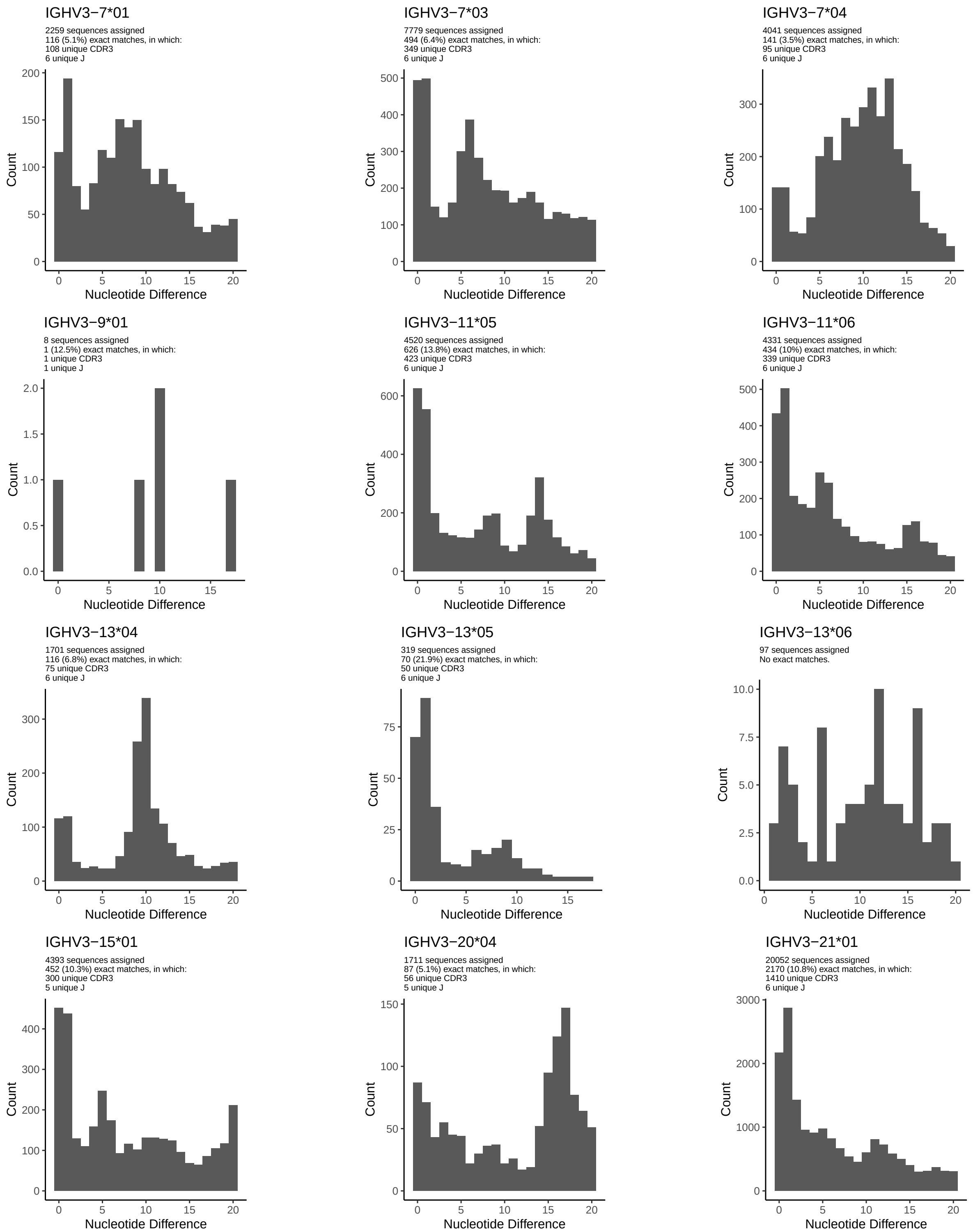

IGHV4−38−2*01_A291G_C300T
Count
CDR3 AA Length (unmutated)

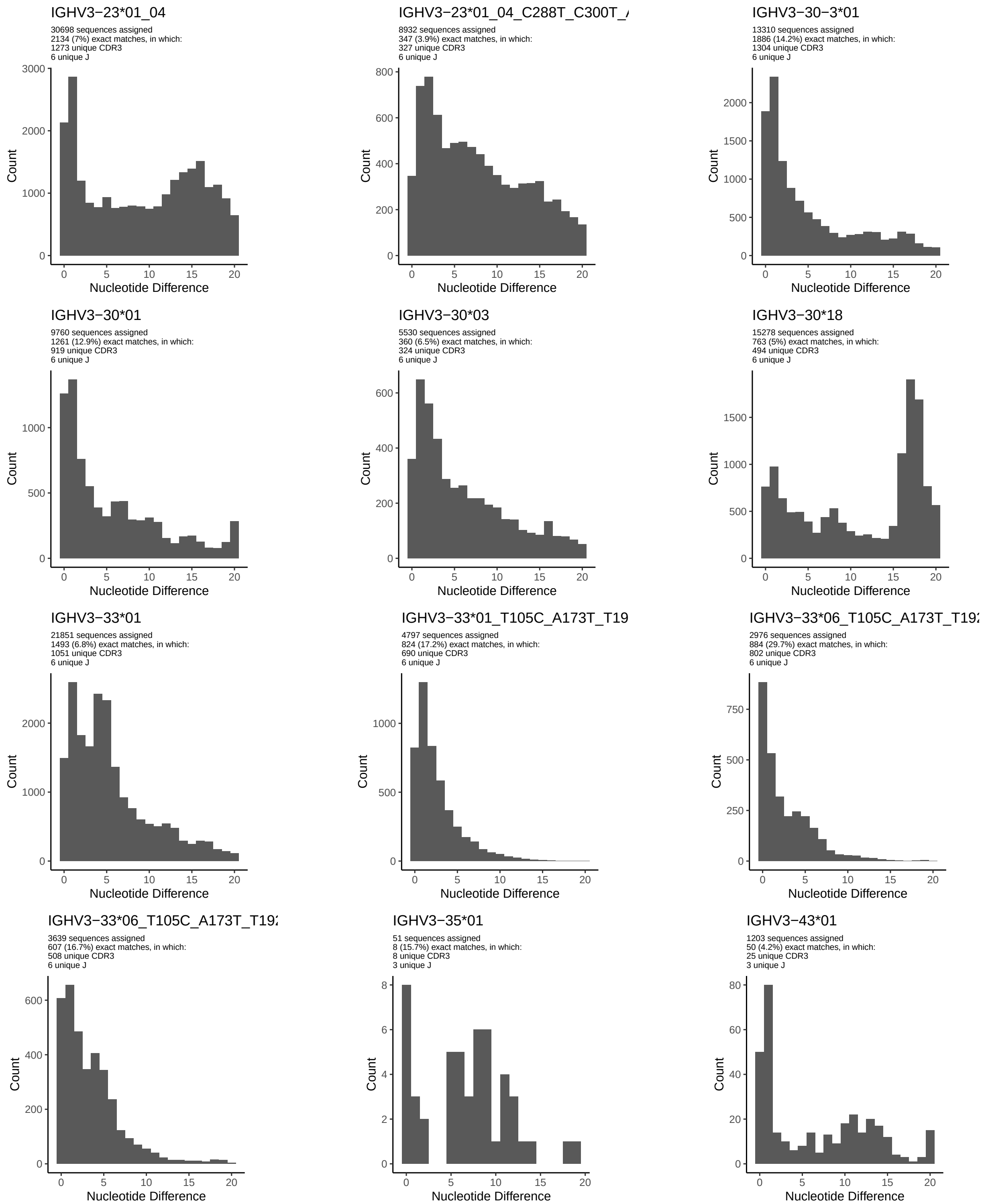

IGHV4−59*12_C300T
Count
CDR3 AA Length (unmutated)

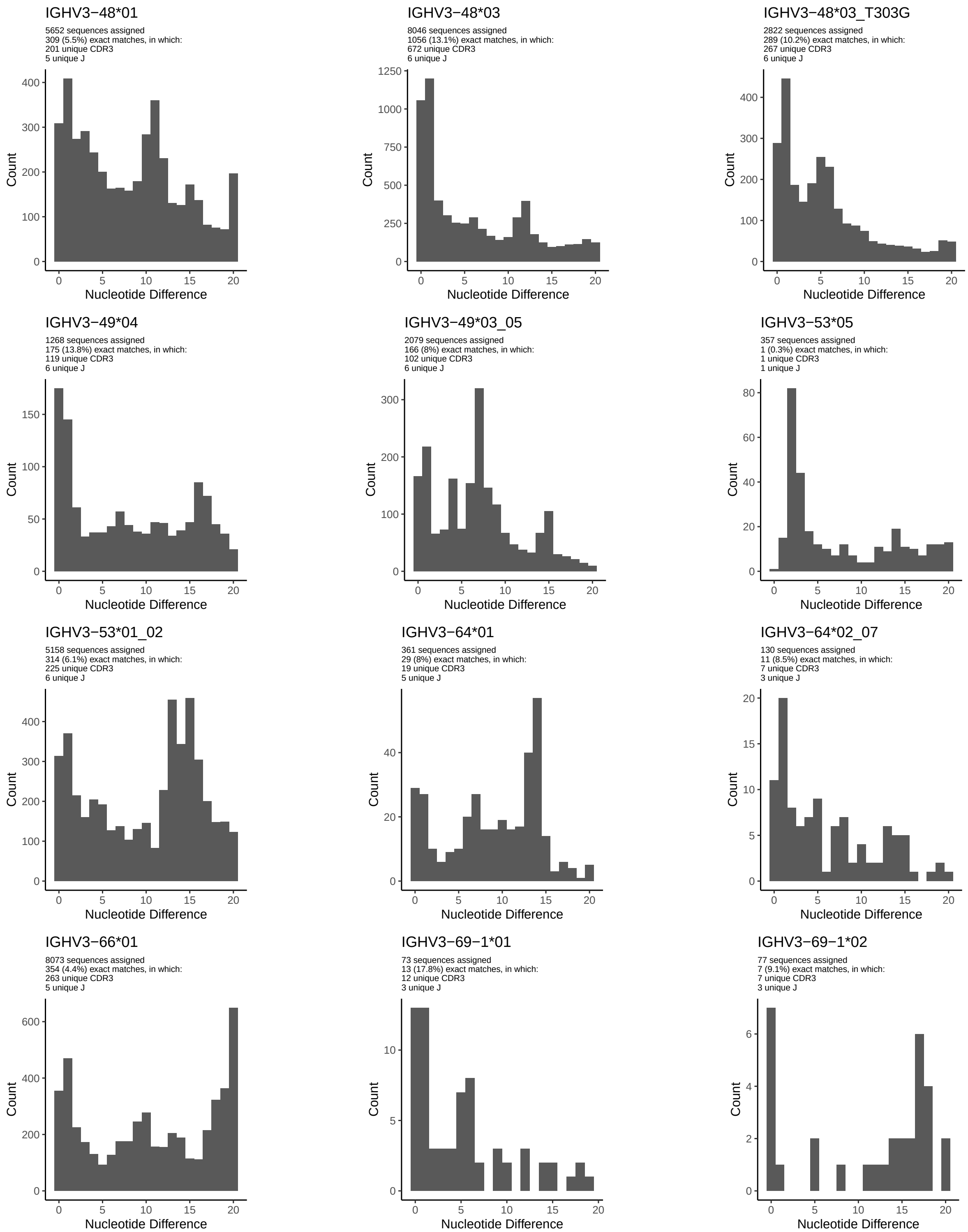
2 Variation from germline, in assignments to each allele

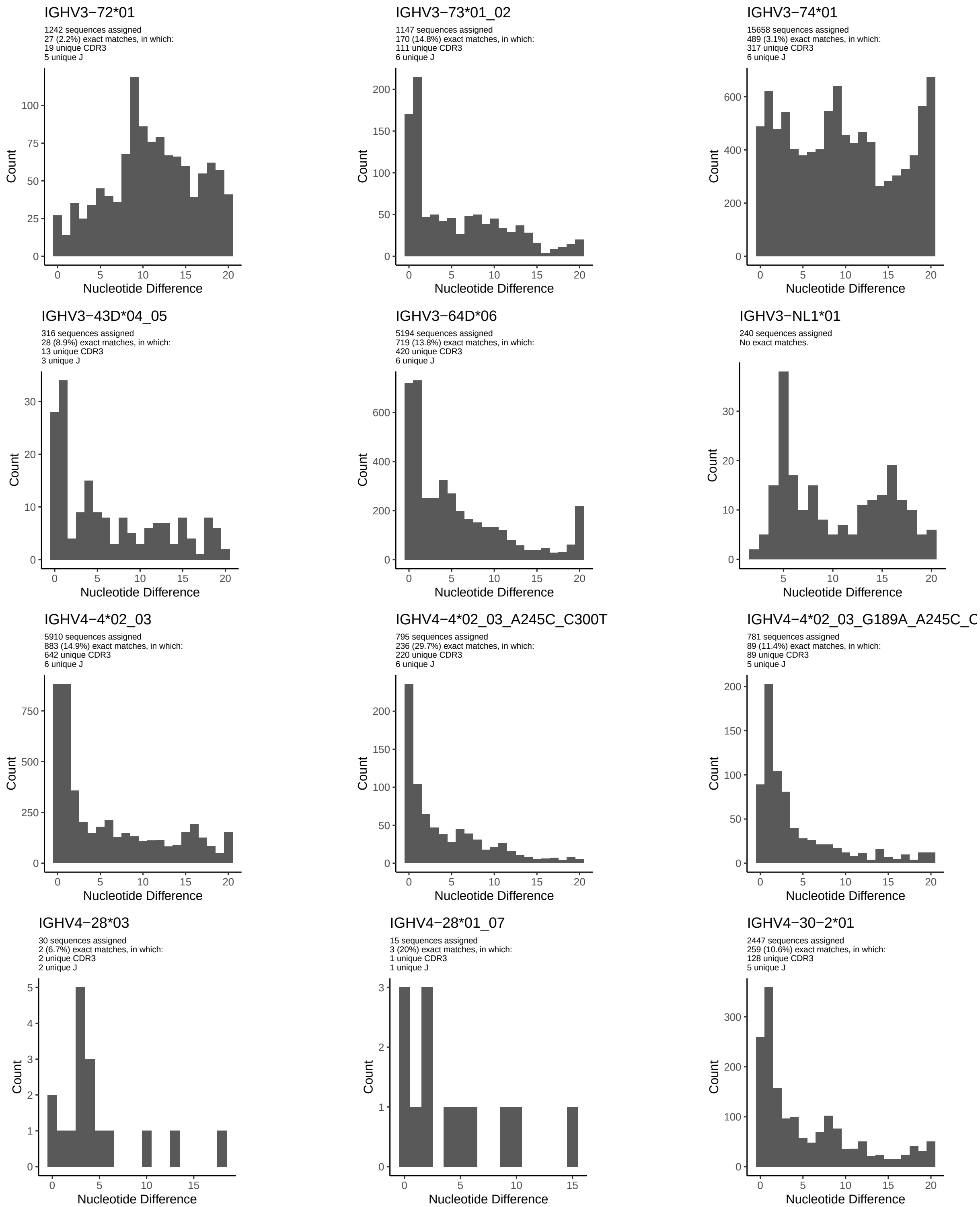


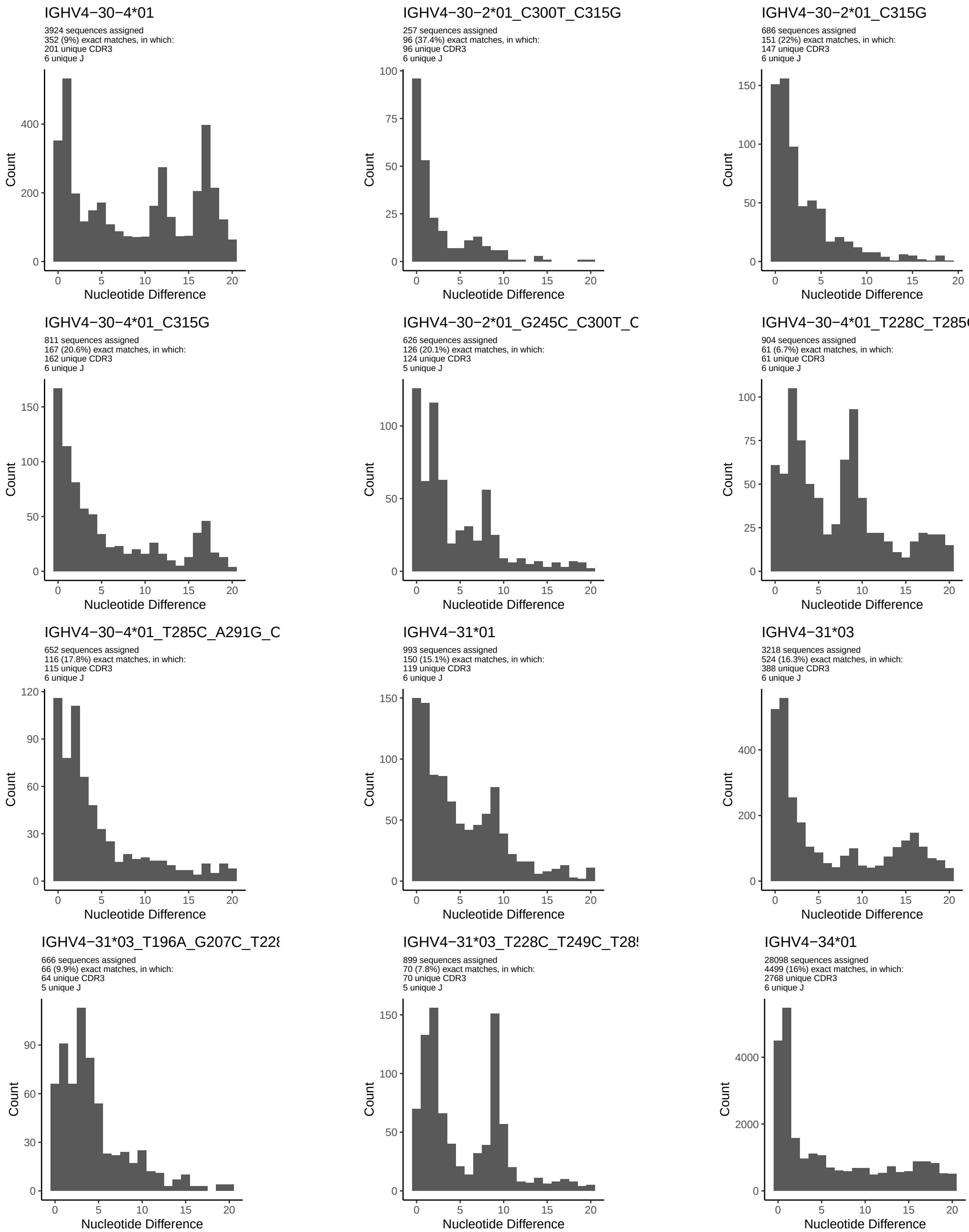

IGHV1−46*01
13243 sequences assigned
1798 (13.6%) exact matches, in which:
1338 unique CDR3
6 unique J
IGHV1−58*02
353 sequences assigned
76 (21.5%) exact matches, in which:
58 unique CDR3
5 unique J
IGHV1−58*01_03
683 sequences assigned
172 (25.2%) exact matches, in which:
117 unique CDR3
6 unique J
IGHV1−69−2*01
22 sequences assigned
1 (4.5%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV1−69*01_12_13
9917 sequences assigned
2363 (23.8%) exact matches, in which:
1777 unique CDR3
6 unique J
IGHV1−69*04_09
7798 sequences assigned
1429 (18.3%) exact matches, in which:
1122 unique CDR3
6 unique J
IGHV1−69*06_14
8125 sequences assigned
1728 (21.3%) exact matches, in which:
1356 unique CDR3
6 unique J
IGHV2−5*02
6970 sequences assigned
239 (3.4%) exact matches, in which:
156 unique CDR3
6 unique J
IGHV2−26*01
2138 sequences assigned
258 (12.1%) exact matches, in which:
160 unique CDR3
6 unique J
IGHV2−70*01
1006 sequences assigned
108 (10.7%) exact matches, in which:
86 unique CDR3
5 unique J
IGHV2−70*15
2192 sequences assigned
122 (5.6%) exact matches, in which:
83 unique CDR3
6 unique J
IGHV2−70D*04
1054 sequences assigned
62 (5.9%) exact matches, in which:
42 unique CDR3
5 unique J
Count
Nucleotide Difference

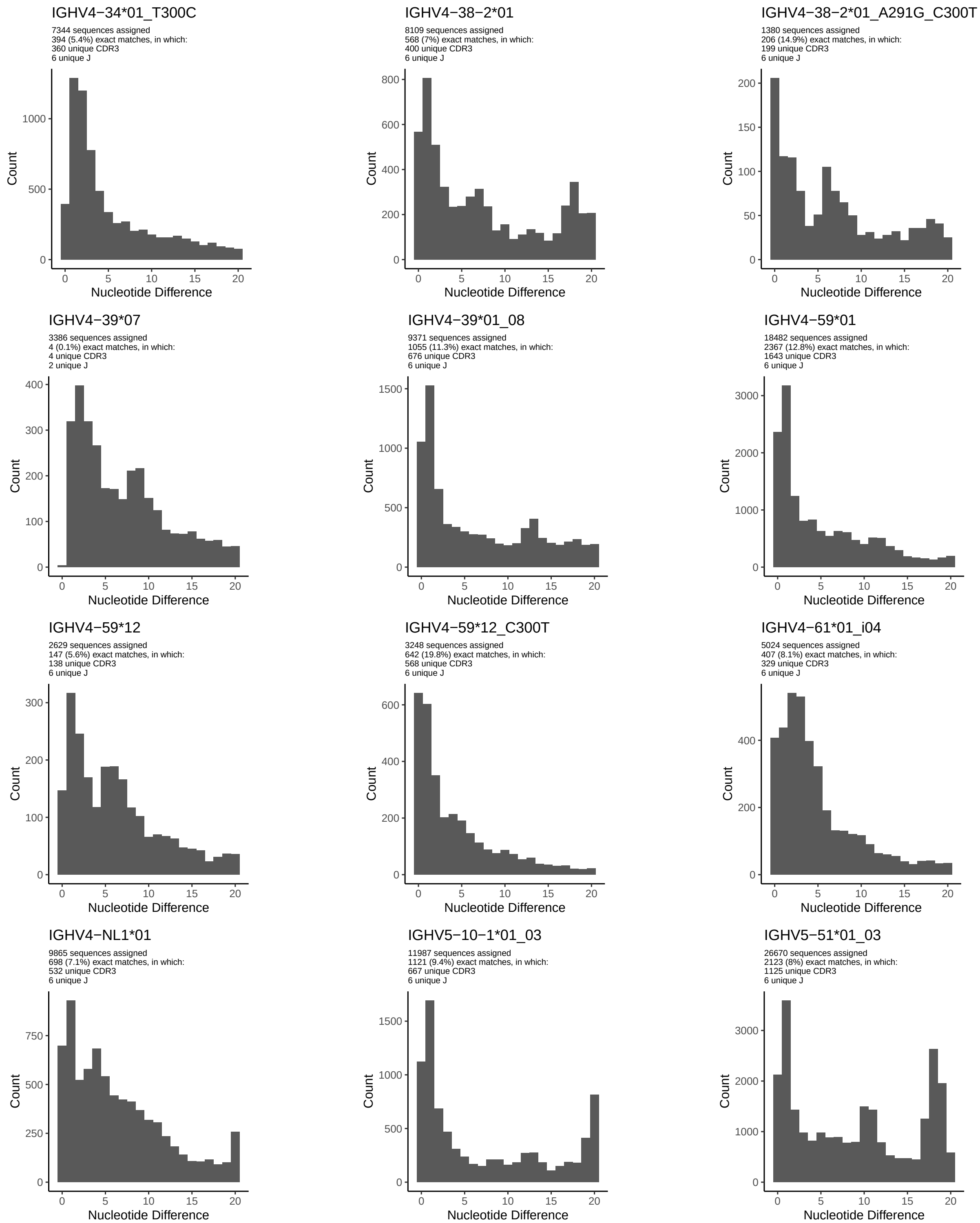

IGHV3−7*01
2259 sequences assigned
116 (5.1%) exact matches, in which:
108 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−7*03
7779 sequences assigned
494 (6.4%) exact matches, in which:
349 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−7*04
4041 sequences assigned
141 (3.5%) exact matches, in which:
95 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−9*01
8 sequences assigned
1 (12.5%) exact matches, in which:
1 unique CDR3
1 unique J
Count
Nucleotide Difference
IGHV3−11*05
4520 sequences assigned
626 (13.8%) exact matches, in which:
423 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−11*06
4331 sequences assigned
434 (10%) exact matches, in which:
339 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−13*04
1701 sequences assigned
116 (6.8%) exact matches, in which:
75 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−13*05
319 sequences assigned
70 (21.9%) exact matches, in which:
50 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−13*06
97 sequences assigned
No exact matches.
Count
Nucleotide Difference
IGHV3−15*01
4393 sequences assigned
452 (10.3%) exact matches, in which:
300 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV3−20*04
1711 sequences assigned
87 (5.1%) exact matches, in which:
56 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV3−21*01
20052 sequences assigned
2170 (10.8%) exact matches, in which:
1410 unique CDR3
6 unique J
Count
Nucleotide Difference

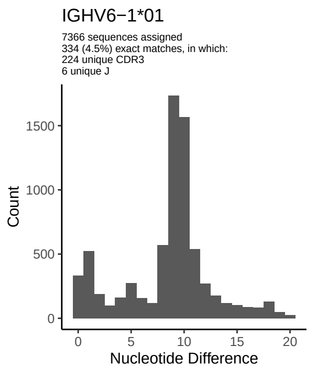

IGHV3−23*01_04
30698 sequences assigned
2134 (7%) exact matches, in which:
1273 unique CDR3
6 unique J
IGHV3−23*01_04_C288T_C300T_
8932 sequences assigned
347 (3.9%) exact matches, in which:
327 unique CDR3
6 unique J
IGHV3−30−3*01
13310 sequences assigned
1886 (14.2%) exact matches, in which:
1304 unique CDR3
6 unique J
IGHV3−30*01
9760 sequences assigned
1261 (12.9%) exact matches, in which:
919 unique CDR3
6 unique J
IGHV3−30*03
5530 sequences assigned
360 (6.5%) exact matches, in which:
324 unique CDR3
6 unique J
IGHV3−30*18
15278 sequences assigned
763 (5%) exact matches, in which:
494 unique CDR3
6 unique J
IGHV3−33*01
21851 sequences assigned
1493 (6.8%) exact matches, in which:
1051 unique CDR3
6 unique J
IGHV3−33*01_T105C_A173T_T19
4797 sequences assigned
824 (17.2%) exact matches, in which:
690 unique CDR3
6 unique J
IGHV3−33*06_T105C_A173T_T19
2976 sequences assigned
884 (29.7%) exact matches, in which:
802 unique CDR3
6 unique J
IGHV3−33*06_T105C_A173T_T19
3639 sequences assigned
607 (16.7%) exact matches, in which:
508 unique CDR3
6 unique J
IGHV3−35*01
51 sequences assigned
8 (15.7%) exact matches, in which:
8 unique CDR3
3 unique J
IGHV3−43*01
1203 sequences assigned
50 (4.2%) exact matches, in which:
25 unique CDR3
3 unique J
Count
Nucleotide Difference

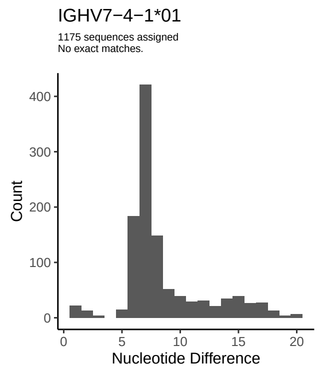

IGHV3−48*01
5652 sequences assigned
309 (5.5%) exact matches, in which:
201 unique CDR3
5 unique J
IGHV3−48*03
8046 sequences assigned
1056 (13.1%) exact matches, in which:
672 unique CDR3
6 unique J
IGHV3−48*03_T303G
2822 sequences assigned
289 (10.2%) exact matches, in which:
267 unique CDR3
6 unique J
IGHV3−49*04
1268 sequences assigned
175 (13.8%) exact matches, in which:
119 unique CDR3
6 unique J
IGHV3−49*03_05
2079 sequences assigned
166 (8%) exact matches, in which:
102 unique CDR3
6 unique J
IGHV3−53*05
357 sequences assigned
1 (0.3%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−53*01_02
5158 sequences assigned
314 (6.1%) exact matches, in which:
225 unique CDR3
6 unique J
IGHV3−64*01
361 sequences assigned
29 (8%) exact matches, in which:
19 unique CDR3
5 unique J
IGHV3−64*02_07
130 sequences assigned
11 (8.5%) exact matches, in which:
7 unique CDR3
3 unique J
IGHV3−66*01
8073 sequences assigned
354 (4.4%) exact matches, in which:
263 unique CDR3
5 unique J
IGHV3−69−1*01
73 sequences assigned
13 (17.8%) exact matches, in which:
12 unique CDR3
3 unique J
IGHV3−69−1*02
77 sequences assigned
7 (9.1%) exact matches, in which:
7 unique CDR3
3 unique J
Count
Nucleotide Difference


IGHV3−72*01
1242 sequences assigned
27 (2.2%) exact matches, in which:
19 unique CDR3
5 unique J
IGHV3−73*01_02
1147 sequences assigned
170 (14.8%) exact matches, in which:
111 unique CDR3
6 unique J
IGHV3−74*01
15658 sequences assigned
489 (3.1%) exact matches, in which:
317 unique CDR3
6 unique J
IGHV3−43D*04_05
316 sequences assigned
28 (8.9%) exact matches, in which:
13 unique CDR3
3 unique J
IGHV3−64D*06
5194 sequences assigned
719 (13.8%) exact matches, in which:
420 unique CDR3
6 unique J
IGHV3−NL1*01
240 sequences assigned
No exact matches.
IGHV4−4*02_03
5910 sequences assigned
883 (14.9%) exact matches, in which:
642 unique CDR3
6 unique J
IGHV4−4*02_03_A245C_C300T
795 sequences assigned
236 (29.7%) exact matches, in which:
220 unique CDR3
6 unique J
IGHV4−4*02_03_G189A_A245C_C
781 sequences assigned
89 (11.4%) exact matches, in which:
89 unique CDR3
5 unique J
IGHV4−28*03
30 sequences assigned
2 (6.7%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV4−28*01_07
15 sequences assigned
3 (20%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV4−30−2*01
2447 sequences assigned
259 (10.6%) exact matches, in which:
128 unique CDR3
5 unique J
Count
Nucleotide Difference

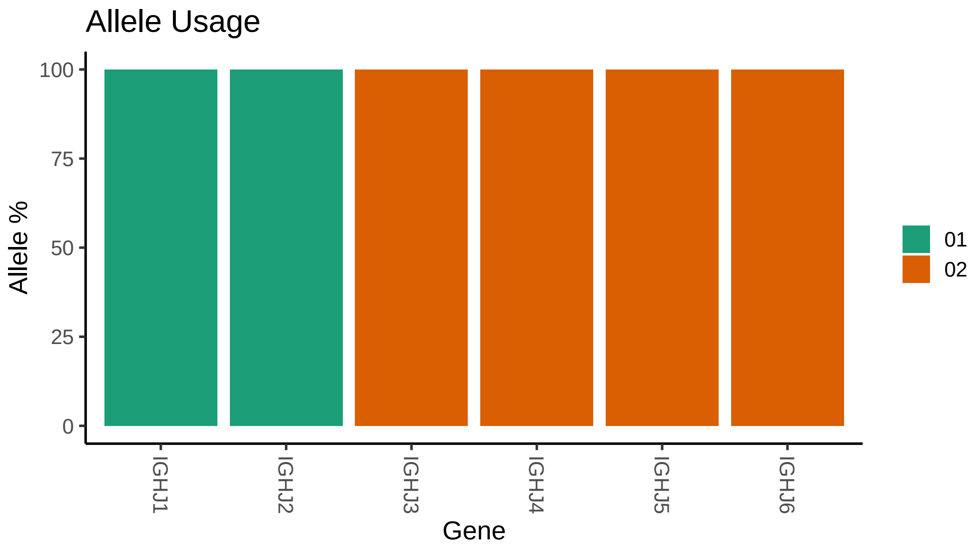

IGHV4−30−4*01
3924 sequences assigned
352 (9%) exact matches, in which:
201 unique CDR3
6 unique J
IGHV4−30−2*01_C300T_C315G
257 sequences assigned
96 (37.4%) exact matches, in which:
96 unique CDR3
6 unique J
IGHV4−30−2*01_C315G
686 sequences assigned
151 (22%) exact matches, in which:
147 unique CDR3
6 unique J
IGHV4−30−4*01_C315G
811 sequences assigned
167 (20.6%) exact matches, in which:
162 unique CDR3
6 unique J
IGHV4−30−2*01_G245C_C300T_C
626 sequences assigned
126 (20.1%) exact matches, in which:
124 unique CDR3
5 unique J
IGHV4−30−4*01_T228C_T285C_A
904 sequences assigned
61 (6.7%) exact matches, in which:
61 unique CDR3
6 unique J
IGHV4−30−4*01_T285C_A291G_C
652 sequences assigned
116 (17.8%) exact matches, in which:
115 unique CDR3
6 unique J
IGHV4−31*01
993 sequences assigned
150 (15.1%) exact matches, in which:
119 unique CDR3
6 unique J
IGHV4−31*03
3218 sequences assigned
524 (16.3%) exact matches, in which:
388 unique CDR3
6 unique J
IGHV4−31*03_T196A_G207C_T22
666 sequences assigned
66 (9.9%) exact matches, in which:
64 unique CDR3
5 unique J
IGHV4−31*03_T228C_T249C_T28
899 sequences assigned
70 (7.8%) exact matches, in which:
70 unique CDR3
5 unique J
IGHV4−34*01
28098 sequences assigned
4499 (16%) exact matches, in which:
2768 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−34*01_T300C
7344 sequences assigned
394 (5.4%) exact matches, in which:
360 unique CDR3
6 unique J
IGHV4−38−2*01
8109 sequences assigned
568 (7%) exact matches, in which:
400 unique CDR3
6 unique J
IGHV4−38−2*01_A291G_C300T
1380 sequences assigned
206 (14.9%) exact matches, in which:
199 unique CDR3
6 unique J
IGHV4−39*07
3386 sequences assigned
4 (0.1%) exact matches, in which:
4 unique CDR3
2 unique J
IGHV4−39*01_08
9371 sequences assigned
1055 (11.3%) exact matches, in which:
676 unique CDR3
6 unique J
IGHV4−59*01
18482 sequences assigned
2367 (12.8%) exact matches, in which:
1643 unique CDR3
6 unique J
IGHV4−59*12
2629 sequences assigned
147 (5.6%) exact matches, in which:
138 unique CDR3
6 unique J
IGHV4−59*12_C300T
3248 sequences assigned
642 (19.8%) exact matches, in which:
568 unique CDR3
6 unique J
IGHV4−61*01_i04
5024 sequences assigned
407 (8.1%) exact matches, in which:
329 unique CDR3
6 unique J
IGHV4−NL1*01
9865 sequences assigned
698 (7.1%) exact matches, in which:
532 unique CDR3
6 unique J
IGHV5−10−1*01_03
11987 sequences assigned
1121 (9.4%) exact matches, in which:
667 unique CDR3
6 unique J
IGHV5−51*01_03
26670 sequences assigned
2123 (8%) exact matches, in which:
1125 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV6−1*01
7366 sequences assigned
334 (4.5%) exact matches, in which:
224 unique CDR3
6 unique J
Count
0
500
1000
1500
0
5
10
15
20
Nucleotide Difference


IGHV7−4−1*01
1175 sequences assigned
No exact matches.
Count
0
100
200
300
400
0
5
10
15
20
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA491287/M64 074/M64 074/M64 074/M64 074/e6/12fcea428e9591aab4
##
## Germline reference file: /misc/work/jenkins/PRJNA491287/M64 074/M64 074/M64 074/M64 074/e6/12fcea428e
##
## Novel allele file: /misc/work/jenkins/PRJNA491287/M64 074/M64 074/M64 074/M64 074/e6/12fcea428e9591aa
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C288T_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 01_T105C_A173T_T192G_C288T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_T105C_A173T_T192G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_T105C_A173T_T192G_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 03_T303G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_G189A_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C300T_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_G245C_C300T_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 4 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_T285C_A291G_C300T_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_T228C_T285C_A291G_C300T_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 31 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 31 03_T228C_T249C_T285C_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 31 03_T196A_G207C_T228C_T249C_T285C_C288T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 01_A291G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
```